Project: Worked examples
Title: Press and shrink fit
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1. Scope and method

Calculates the interface pressure, stresses, assembly force and friction torque of a shaft pressed or shrunk into a hub, and the shrink temperature. It uses the Lamé thick-cylinder relations and ignores end effects.

WHAT IT CALCULATES	RELATION
Radial interference	$\delta_r = \delta/2$
Interface pressure, the hub and the shaft as thick cylinders	$p_{int} = \delta_r / [\ (R/E_h)\cdot((r_o^2 + R^2)/(r_o^2 - R^2) + \nu_h) + (R/E_s)\cdot((R^2 + r_i^2)/(R^2 - r_i^2) - \nu_s)\ ]$
Stresses at the hub bore	$\sigma_\theta = p_{int}\cdot(r_o^2 + R^2)/(r_o^2 - R^2)$; $\sigma_r = -p_{int}$
Stresses at the shaft surface	$\sigma_\theta = -p_{int}\cdot(R^2 + r_i^2)/(R^2 - r_i^2)$; $\sigma_r = -p_{int}$
Stress at the bore of a hollow shaft	$\sigma_\theta = -2\cdot p_{int}\cdot R^2/(R^2 - r_i^2)$; $\sigma_r = 0$
Surface smoothing on assembly (when Rz is entered)	$\delta_{eff} = \max(\delta - 0.8\cdot(Rz_h + Rz_s), 0)$, $p_{eff} = p_{int}\cdot\delta_{eff}/\delta$
Axial force and torque the fit holds by friction, on p_{eff}	$F = \mu\cdot p_{eff}\cdot 2\pi R\cdot L$; $T = \mu\cdot p_{eff}\cdot 2\pi R^2\cdot L$. The stresses stay at the nominal p_{int} (the larger interference of the fit)
Temperature difference to assemble by heating the hub, with assembly clearance c_a	$\Delta T = (\delta + c_a)/(\alpha\cdot 2R)$

Basis: Press and shrink fits; stresses in pressurised cylinders[1](§3-16, §3-17, 11th ed.); Thick-walled cylinders under pressure[2](Ch. 13, Table 13.5, 8th ed.).

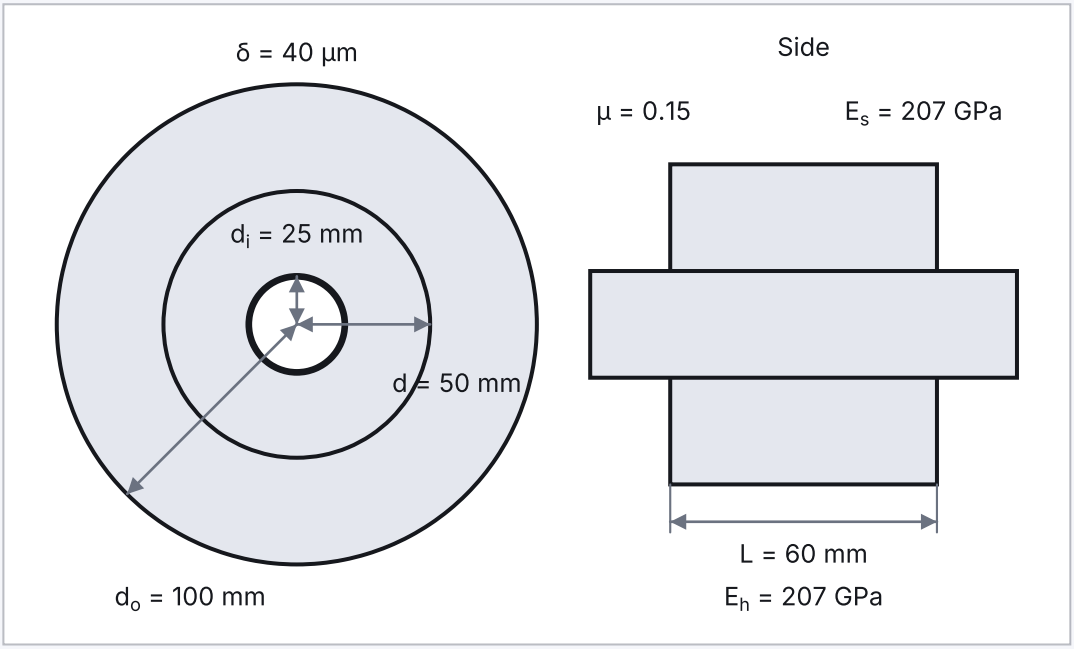


Figure 1 — A hub shrunk onto a shaft: section across the interface, and the engagement length beside it. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d	Interface (nominal) diameter	50	mm	drawing: DRG-HUB-04 rev C	1 … 1000 mm
δ	Diametral interference	40	µm	drawing: DRG-HUB-04 rev C	1 … 500 µm
d_o	Hub outer diameter	100	mm	drawing: DRG-HUB-04 rev C	1 … 2000 mm
d_i	Shaft bore diameter (0 for a solid shaft)	25	mm	drawing: DRG-SHAFT-13 rev A	0 … 1000 mm
L	Engagement length	60	mm	drawing: DRG-HUB-04 rev C	1 … 2000 mm
E_h	Hub Young's modulus	207	GPa	spec: steel	10 … 450 GPa
ν_h	Hub Poisson's ratio	0.292		spec: steel	0.1 … 0.45
E_s	Shaft Young's modulus	207	GPa	spec: steel	10 … 450 GPa
ν_s	Shaft Poisson's ratio	0.292		spec: steel	0.1 … 0.45
$S_{y,h}$	Hub yield strength	355	MPa	spec: EN 10025 S355	50 … 2000 MPa
$S_{y,s}$	Shaft yield strength	355	MPa	spec: EN 10025 S355	50 … 2000 MPa
α_h	Hub thermal expansion coefficient	0.000012	1/K	spec: steel	0.000005 … 0.00003 1/K
μ	Coefficient of friction at the interface	0.15		assumption: Dry steel on steel.	0.05 … 0.4
c_a	Assembly clearance wanted for shrink fitting	20	µm	assumption: Assembly clearance.	0 … 500 µm

3. Assumptions

A1. Both parts are long cylinders in plane stress, and the interface pressure is uniform along the engagement. (The Lamé thick-cylinder relations apply, and end effects are neglected.)
A2. The interference is the nominal value entered. (Use the minimum interference of the fit for the torque and the force, and the maximum for the stresses.)
A3. Surface roughness smooths on assembly and takes about $0.8 \cdot (Rz_h + Rz_s)$ off the interference when Rz is entered. (DIN 7190-1 gives the allowance; enter Rz to apply it to the torque and the force.)
A4. The friction coefficient is as entered and the same for assembly and for torque transmission. (Enter it with its source, because it dominates the force and torque results.)
A5. Yield is checked by von Mises stress at the hub bore, the shaft surface and a hollow shaft's bore. (The axial stress is zero, and the larger of the two shaft values governs.)
A6. μ (Coefficient of friction at the interface) is taken as 0.15. (Dry steel on steel.)
A7. c_a (Assembly clearance wanted for shrink fitting) is taken as 20 µm. (Assembly clearance.)
A8. No surface smoothing is applied, so the torque needs the minimum interference of the fit and the stresses the maximum. (Roughness takes about $0.8 \cdot (Rz_h + Rz_s)$ off the interference, so enter Rz_h and Rz_s to apply it (DIN 7190-1).)

4. Working

[15] $R = d / 2$

$= 50\ \text{mm} / 2$

$= 25\ \text{mm}$

Interface radius

[16] $r_o = d_o / 2$

$= 100\ \text{mm} / 2$

$= 50\ \text{mm}$

Hub outer radius

[17] $r_i = d_i / 2$

$= 25\ \text{mm} / 2$

$= 12.5\ \text{mm}$

Shaft bore radius

[18] $\delta_r = \delta / 2$

$= 40\ \mu\text{m} / 2$

$= 20\ \mu\text{m}$

Radial interference

[19] $C_h = (R / E_h) * ((r_o^2 + R^2) / (r_o^2 - R^2) + \nu_h)$ [1](§3-16)

$= 25\ \text{mm} / 207\ \text{GPa} \times (((50\ \text{mm})^2 + (25\ \text{mm})^2) / ((50\ \text{mm})^2 - (25\ \text{mm})^2) + 0.292)$

$= 0.23655\ \text{mm/GPa}$

Hub compliance

[20] $C_s = (R / E_s) * ((R^2 + r_i^2) / (R^2 - r_i^2) - \nu_s)$ [1](§3-16)

$= 25\ \text{mm} / 207\ \text{GPa} \times (((25\ \text{mm})^2 + (12.5\ \text{mm})^2) / ((25\ \text{mm})^2 - (12.5\ \text{mm})^2) - 0.292)$

$= 0.16602\ \text{mm/GPa}$

Shaft compliance

[21] $p_{int} = \delta_r / (C_h + C_s)$ [1](eq. 3-56)

$= 20\ \mu\text{m} / (0.23655\ \text{mm/GPa} + 0.16602\ \text{mm/GPa})$

$= 49.68\ \text{MPa}$

Interface pressure

[22] $\sigma_{t,hub} = p_{int} * (r_o^2 + R^2) / (r_o^2 - R^2)$ [1](eq. 3-57)

$= 49.68\ \text{MPa} \times ((50\ \text{mm})^2 + (25\ \text{mm})^2) / ((50\ \text{mm})^2 - (25\ \text{mm})^2)$

$= 82.8\ \text{MPa}$

Hub hoop stress at the bore

[23] $\sigma_{r,hub} = -p_{int}$

$= -49.68\ \text{MPa}$

$= -49.68\ \text{MPa}$

Hub radial stress at the bore

[24] $\sigma_{t,shaft} = -p_{int} * (R^2 + r_i^2) / (R^2 - r_i^2)$ [1](eq. 3-58)

$= -49.68\ \text{MPa} \times ((25\ \text{mm})^2 + (12.5\ \text{mm})^2) / ((25\ \text{mm})^2 - (12.5\ \text{mm})^2)$

$= -82.8\ \text{MPa}$

Shaft hoop stress at the surface

[25] $\sigma_{r,shaft} = -p_{int}$

$= -49.68\ \text{MPa}$

$= -49.68\ \text{MPa}$

Shaft radial stress at the surface

[26] $\sigma_{vm,hub} = \text{sqrt}(\sigma_{t,hub}^2 - \sigma_{t,hub} * \sigma_{r,hub} + \sigma_{r,hub}^2)$

$= \text{sqrt}((82.8\ \text{MPa})^2 - 82.8\ \text{MPa} \times (-49.68\ \text{MPa}) + (-49.68\ \text{MPa})^2)$

$= 115.92\ \text{MPa}$

Hub von Mises stress at the bore

[27] $\sigma_{vm,shaft} = \text{sqrt}(\sigma_{t,shaft}^2 - \sigma_{t,shaft} * \sigma_{r,shaft} + \sigma_{r,shaft}^2)$

$= \text{sqrt}((-82.8\ \text{MPa})^2 - (-82.8\ \text{MPa}) \times (-49.68\ \text{MPa}) + (-49.68\ \text{MPa})^2)$

$= 72.183\ \text{MPa}$

Shaft von Mises stress at the surface

[28] $\sigma_{t,shaft,bore}$ = -2 * p_{int} * R^2 / (R^2 - r_i^2) [3](cylinder)
= -2 × 49.68 MPa × (25 mm)^2 / ((25 mm)^2 - (12.5 mm)^2)
= -132.48 MPa
Shaft hoop stress at the shaft bore (hollow shaft; the radial stress is zero there)
[29] $\sigma_{vm,shaft,bore}$ = abs($\sigma_{t,shaft,bore}$)
= abs(-132.48 MPa)
= 132.48 MPa
Shaft von Mises stress at the shaft bore (hollow shaft: hoop stress alone)
[30] $\sigma_{vm,shaft,max}$ = max($\sigma_{vm,shaft}$, $\sigma_{vm,shaft,bore}$)
= max(72.183 MPa, 132.48 MPa)
= 132.48 MPa
Largest shaft von Mises stress, surface or bore
[31] δ_{eff} = δ
= 40 µm
= 40 µm
Effective interference (no roughness entered: the nominal interference)
[32] p_{eff} = p_{int} * δ_{eff} / δ
= 49.68 MPa × 40 µm / 40 µm
= 49.68 MPa
Interface pressure that grips, at the effective interference
[33] F_a = µ * p_{eff} * 2 * pi * R * L
= 0.15 × 49.68 MPa × 2 × π × 25 mm × 60 mm
= 70.233 kN
Axial assembly (press) force
[34] T = µ * p_{eff} * 2 * pi * R^2 * L
= 0.15 × 49.68 MPa × 2 × π × (25 mm)^2 × 60 mm
= 1.7558 kN·m
Torque capacity by friction
[35] dT = (δ + c_a) / (α_h * d)
= (40 µm + 20 µm) / (0.000012 1/K × 50 mm)
= 100 K
Hub temperature rise for shrink fitting

5. Results

SYMBOL	VALUE	DESCRIPTION
p_{int}	49.68 MPa	Interface pressure
T	1.756 kN·m	Torque capacity
F_a	70.23 kN	Assembly force
dT	100 K	Shrink-fit temperature rise
$\sigma_{vm,hub}$	115.9 MPa	Hub von Mises stress
$\sigma_{vm,shaft,max}$	132.5 MPa	Largest shaft von Mises stress (surface or bore) (governing)
$\sigma_{t,hub}$	82.80 MPa	Hub hoop stress at bore
$\sigma_{t,shaft}$	-82.80 MPa	Shaft hoop stress at surface
$\sigma_{vm,shaft}$	72.18 MPa	Shaft von Mises stress at the surface
$\sigma_{vm,shaft,bore}$	132.5 MPa	Shaft von Mises stress at the shaft bore
δ_{eff}	40 µm	Effective interference after smoothing
p_{eff}	49.68 MPa	Gripping interface pressure

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Largest shaft von Mises stress against yield strength	132.5 MPa	355 MPa	0.37	
PASS	Hub von Mises stress against yield strength	115.9 MPa	355 MPa	0.33	

7. Applicability

None.

8. Verification

Template mech.press-fit@1.3.0 Content hash ff1d6e619181e260 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
2. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
3. Lamé, external pressure on a hollow.